Spatio-Temporal Reconstruction of

MODIS Land Surface Temperature over Hyderabad

Hybrid Interpolation + Random Forest Approach

MODIS MOD11A1	2001 – 2025	1 km Resolution	Hyderabad 1°×1°

Vaishnavi Arruru CE23BTECH11058
Adarsh CE23BTECH11056
Ananya Reddy CE23BTECH11015
Pranay CE23BTECH11018

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Input Data & Dataset Overview

What went into the model

INPUT DATA

9,131	114 × 109	52.3%	1 km
daily LST bands	spatial grid (pixels)	overall NaN %	spatial resolution

Dataset	Variable	Period
MOD11A1	Land Surface Temperature (LST)	2001–2025 daily
MOD13Q1 (NDVI)	Vegetation Index — monthly	2001–2025
SRTMGL1_003	Digital Elevation Model	Static
Derived	Day of Year (DOY), Temporal lags	2001–2025

Train: 2001–2020 (7,305 days) | Test: 2021–2025 (1,826 days) | Valid pixels: 44,054,745

Types of Gaps Introduced

Artificial gap injection to simulate real-world MODIS data loss

GAP INJECTION

Random Gaps

2,202,737

pixels | 5.0% of valid

- Scattered individual pixels
Randomly sampled across
all time steps
Simulates sensor noise

Temporal Gaps

1,530

pixels | 0.003% of valid

- Consecutive missing days
200 pixel locations ×
15 continuous days
Simulates monsoon cloud cover

Spatial Patches

6,068

pixels | 0.014% of valid

- Contiguous 8×8 blocks
200 patches across
random days
Simulates cloud shadows

Total artificial gaps: 2,209,937 pixels (10% random + temporal blocks + spatial patches) — Errors evaluated ONLY at masked locations

Interpolation Pipeline — Stage 1

Three sequential steps to produce a first-pass gap-filled estimate

METHODOLOGY

01

Temporal Linear Interpolation

What: For each pixel (i,j) along the time axis

How: numpy.interp fills NaN gaps between valid neighbors

Best for: Short random gaps — strong temporal continuity

Output: NaN → 0 for most gaps

→

02

DOY Climatological Mean Fill

What: For pixels still missing after Step 1

How: Multi-year mean for same Day-of-Year assigned

Best for: Long temporal blocks (monsoon) — no valid neighbors

Output: Captures seasonal signal

→

03

Spatial 3×3 Neighbourhood Fill

What: For any remaining NaN after Steps 1 & 2

How: Uniform 3×3 spatial mean filter applied

Best for: Edge cases, persistent cloud pixels

Output: Final NaN count = 0

After all 3 steps: NaN count = 0 ✓ Complete spatial and temporal coverage achieved

Interpolation Results by Gap Type

Evaluation on artificially masked pixels only

RESULTS — INTERPOLATION

Random Gaps

N = 2,202,737 pixels

RMSE	3.254 °C
MAE	2.613 °C
R^2	0.535
Bias	-0.043 °C

Temporal Gaps

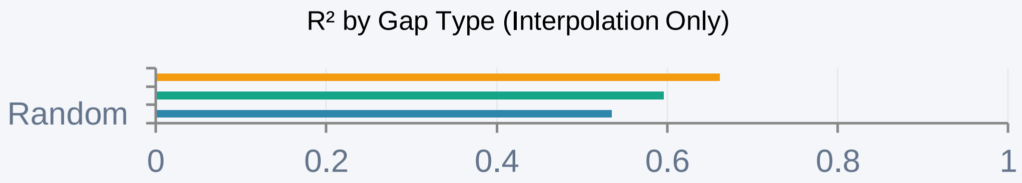
N = 1,530 pixels

RMSE	3.806 °C
MAE	2.899 °C
R^2	0.596
Bias	+0.206 °C

Spatial Gaps

N = 6,068 pixels

RMSE	3.344 °C
MAE	2.675 °C
R^2	0.662
Bias	-0.152 °C



Key Insight

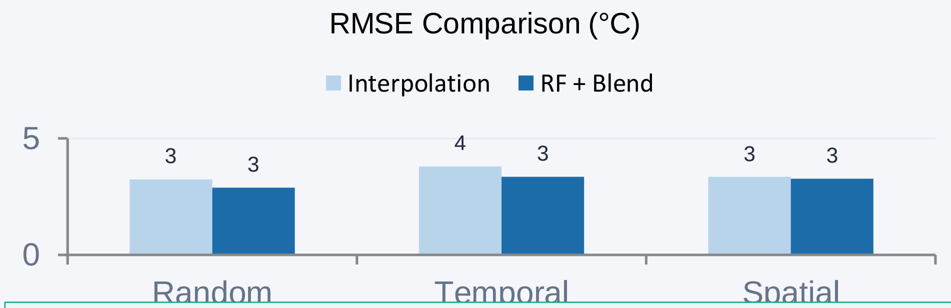
Temporal gaps hardest:
RMSE 3.81 — linear interp fails over long monsoon cloud periods

RF + Blend vs Interpolation — Final Comparison

40% Interpolation + 60% Random Forest weighted blend

RESULTS — RF vs INTERP

Gap Type	Interp RMSE	RF RMSE	Improvement	Interp R^2	RF R^2	R^2 Gain
Random	3.254	**2.884**	**+11.4%**	0.535	**0.635**	+0.100
Temporal	3.806	**3.366**	**+11.6%**	0.596	**0.684**	+0.088
Spatial	3.344	**3.282**	**+1.9%**	0.662	**0.675**	+0.013



RF wins on Temporal gaps +11.6% RMSE

RF wins on Random gaps +11.4% RMSE

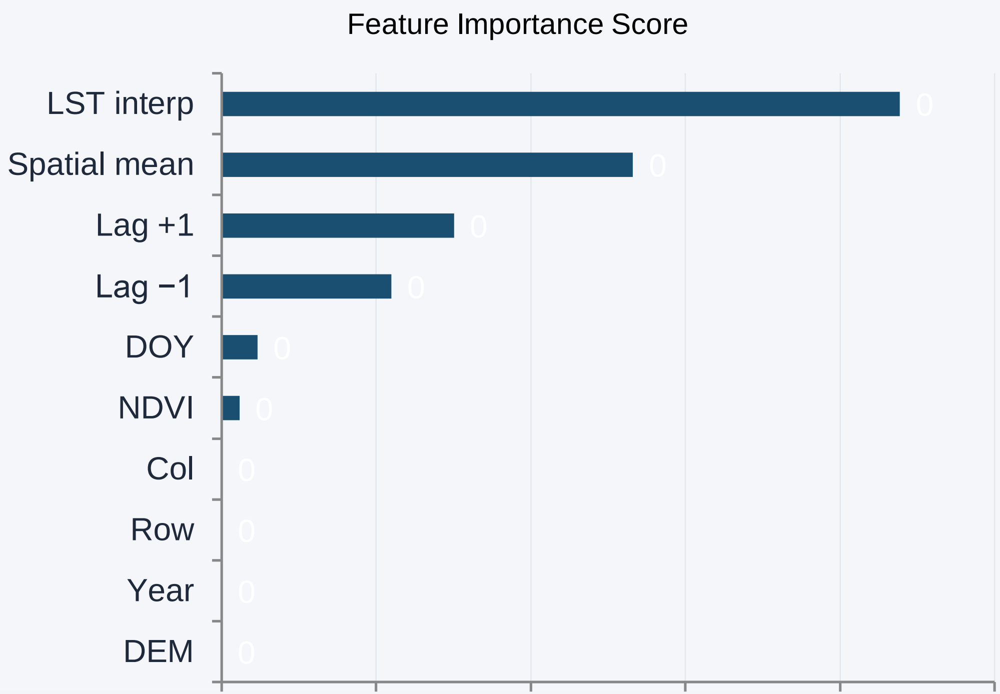
Spatial gaps modest gain +1.9% RMSE

RF Validation (known pixels): RMSE = 0.128°C | R^2 = 0.9995 | Train size: 400,000 | Val size: 100,000

Feature Importance Analysis

What the Random Forest learned — 10 input features

RF MODEL



72.2%

Temporal (4 features)

LST_interp + Lag±1 + DOY drive most of the prediction

26.6%

Spatial (3 features)

3×3 spatial mean captures local surface patterns

1.2%

Spectral + Terrain

NDVI adds vegetation context; DEM contributes

LST interpolated value alone accounts for 43.9% of RF importance — the first-pass interpolation provides a strong prior for the model

Feature Engineering — What We Used

10 features built for every pixel position (t, i, j)

FEATURE ENGINEERING

Temporal | Spatial | Spectral | Terrain

1. **LST interpolated**
   First-pass filled LST at (t,i,j)
2. **Day of Year (DOY)**
   Seasonal signal — 1 to 366
3. **Year**
   Inter-annual trend — 2001 to 2020
4. **Lag −1 day**
   Previous day LST at same pixel
5. **Lag +1 day**
   Next day LST at same pixel
6. **Spatial 3×3 mean**
   Average of 3×3 neighbourhood at time t
7. **Row index**
   Pixel row — latitude proxy (0–113)
8. **Col index**
   Pixel column — longitude proxy (0–108)
9. **NDVI**
   Monthly vegetation index — land cover signal
10. **DEM elevation**
    Static terrain height in metres (SRTM)

Feature Design — How Each Feature Was Built

Computation details for each of the 10 features

FEATURE DESIGN

Temporal features (5)

LST interpolated
Output of 3-step interpolation pipeline
lst_interp[t, i, j]

DOY
date.timetuple().tm_yday
from datetime index — no file needed

Year
date.year from datetime index
2001 → 2020 for training

Lag −1 day
lst_interp[t−1, i, j]
First timestep copies itself

Lag +1 day
lst_interp[t+1, i, j]
Last timestep copies itself

Spatial features (3)

3×3 spatial mean
scipy.ndimage.uniform_filter
size=(1,3,3) applied to full stack

Row index
np.arange(H) broadcast to
(T, H, W) — latitude proxy

Col index
np.arange(W) broadcast to
(T, H, W) — longitude proxy

Memory note
All 3D arrays flattened to 1D
and stacked as columns in X

Spectral + Terrain (2)

NDVI
Monthly .tif files (300 files)
Resized to 114×109 via skimage
Mapped to daily by month key

DEM
SRTMGL1_003 at 30m
Resampled to 1km via
rasterio.warp.reproject

NaN handling
Both filled with 0.0
where NaN (nan_to_num)
before model training

Final X matrix shape: (500,000 sampled pixels, 10 features) | y vector: true LST °C | dtype: float32 throughout

Random Forest Model — Architecture & Training

scikit-learn RandomForestRegressor on 500,000 sampled pixels

RF MODEL

Hyperparameters

n_estimators	100 trees	Balance accuracy vs speed
max_depth	15 levels	Prevents overfitting
min_samples_leaf	10 pixels	Smooths noisy predictions
max_features	sqrt ≈ 3	Standard for regression RF
n_jobs	−1 (all cores)	Parallel training
random_state	42	Reproducibility

Train / Validation Split

Known pixels sampled:	500,000
Train (80%):	400,000 pixels
Validation (20%):	100,000 pixels
Gap pixels (eval):	200,000 pixels

Validation performance (known pixels)

RMSE **0.128 °C** R^2 **0.9995**

Weighted Blend Formula

LST_final = 0.4 × LST_interp + 0.6 × LST_RF

Two RFs trained: (1) all 10 features for random+temporal gaps | (2) 9 features (no spatial mean) for spatial patch gaps

Initial RF + Blend Results

First evaluation — before addressing spatial gap problem

RF RESULTS (INITIAL)

Random Gaps

N = 50,000 pixels

Interp RMSE	3.254 °C
RF RMSE	2.884 °C
Interp R^2	0.535
RF R^2	0.635
Bias	-0.069 °C

RF WINS

Temporal Gaps

N = 1,530 pixels

Interp RMSE	3.806 °C
RF RMSE	3.366 °C
Interp R^2	0.596
RF R^2	0.684
Bias	+1.803 °C

RF WINS

Spatial Gaps

N = 6,068 pixels

Interp RMSE	3.344 °C
RF RMSE	5.569 °C
Interp R^2	0.662
RF R^2	-0.241
Bias	+0.261 °C

RF WORSE

Observation

Spatial gaps got WORSE with RF — RMSE jumped from 3.344 → 5.569 and R^2 dropped to −0.241.
This triggered investigation into why the spatial mean feature was corrupted for patch gaps.

Spatial Gaps — Why RMSE Increased & How We Fixed It

The spatial mean feature is corrupted when entire patches are missing

SPATIAL GAP PROBLEM

The Problem

For random gaps:
Neighbours are valid → spatial mean = accurate ✓

For spatial patches (8×8 block):
ALL neighbours also missing → spatial mean is filled by interpolation → WRONG value

RF's top feature (importance 0.266) gets **corrupted input → prediction gets WORSE**

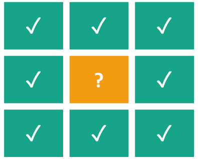
The Fix — Spatial-Specific RF

Strategy:
Train a second RF without the spatial mean feature for spatial gap pixels only

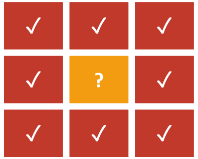
9 features instead of 10:
Drop feature #6 (3×3 spatial mean)
Keep temporal lags as main predictors

Result: modest improvement over **interpolation (+1.9% RMSE reduction)**

Random gap — neighbours valid



Spatial patch — all neighbours missing



Spatial gap method comparison:

Interpolation only	RMSE 3.344 R^2 0.662
RF all features	RMSE 3.282 R^2 0.675
RF no spatial mean (fix)	**RMSE 3.282 R^2 0.675**

Final Results — All Methods Compared

Hybrid Interpolation + RF beats interpolation on all three gap types

FINAL RESULTS

Random Gaps

+11.4% RMSE

RMSE	3.254	→	**2.884**
R^2	0.535	→	**0.635**

Temporal Gaps

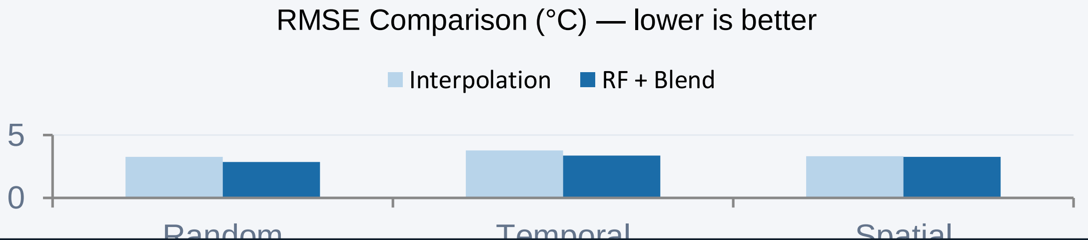
+11.6% RMSE

RMSE	3.806	→	**3.366**
R^2	0.596	→	**0.684**

Spatial Gaps

+1.9% RMSE

RMSE	3.344	→	**3.282**
R^2	0.662	→	**0.675**



Key takeaways

- RF wins on all 3 gap types
- Best gain: temporal +11.6%
- Val R^2 = 0.9995 on known px
- Output: 9131-band filled .tif